

MUZZLE-LOADING ARTILLERY

In the 19th century, infantrymen firing rifled small arms were able to achieve greater range than smoothbore cannon, so attempts were made to apply rifling to field artillery. The first rifled cannon were developed in the 1840s by pioneers including Giovanni Cavalli of Italy, Martin von Wahrendorff of Sweden, and Joseph Whitworth and Charles Lancaster of Britain. Some of these early rifled field guns – such as the rifled Whitworth 12-pounder, which saw service in the American Civil War – were breech- rather than muzzle-loaders. Breech-loaders permitted a higher rate of fire and were generally safer to use, but muzzle-loaders were cheaper to manufacture and required less expensive ammunition. As a result, muzzle-loaders continued to be popular for military use.

► CHINESE 18-POUNDER

Date	1830
Origin	China
Weight	2.5 tonnes (2.75 tons)
Length	3.2m (10½ft)
Calibre	5.25in

With inscriptions in Chinese on top of its breech, this 18-pounder is 48cm (19in) in diameter. It is mounted on a Russian wrought- and cast-iron carriage, which dates to 1853.



Muzzle diameter
41cm (16in)

Wrought-
and cast-iron
garrison carriage

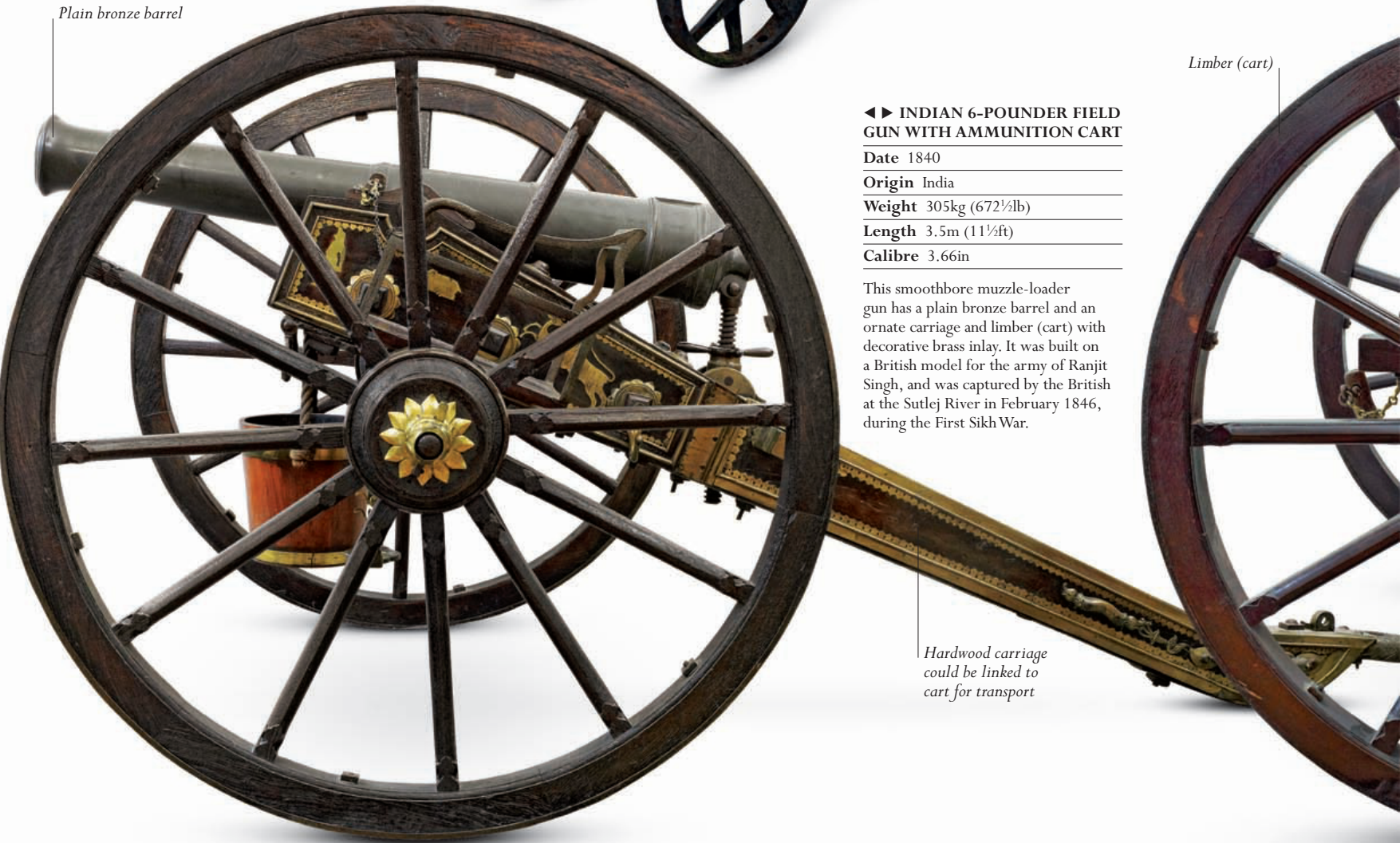


Later
replacement
carriage

▲ RUSSIAN LICORNE

Date	1793
Origin	Russia
Weight	2.76 tonnes (3 tons)
Length	2.8m (9ft)
Calibre	205mm

This gun, which saw action in the Crimean War (1853–56), could fire horizontally or at an elevated trajectory. The licorne was a development of the howitzer, with a powder chamber in the shape of a cone. It could shoot shells as well as cannonballs.



Plain bronze barrel

Limber (cart)

◀ ▶ INDIAN 6-POUNDER FIELD GUN WITH AMMUNITION CART

Date	1840
Origin	India
Weight	305kg (672½lb)
Length	3.5m (11½ft)
Calibre	3.66in

This smoothbore muzzle-loader gun has a plain bronze barrel and an ornate carriage and limber (cart) with decorative brass inlay. It was built on a British model for the army of Ranjit Singh, and was captured by the British at the Sutlej River in February 1846, during the First Sikh War.

Hardwood carriage
could be linked to
cart for transport